

Gebze Technical University

Dep. of Mechanical Engineering

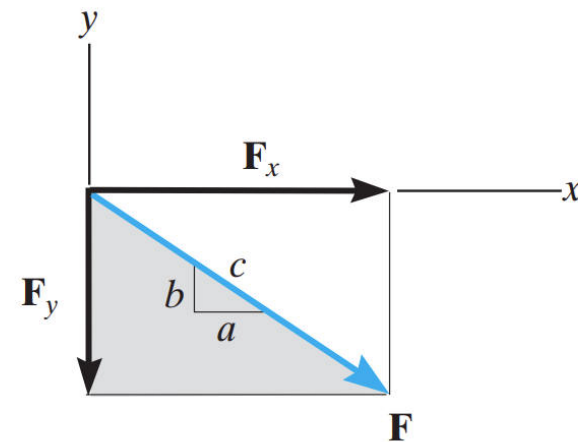
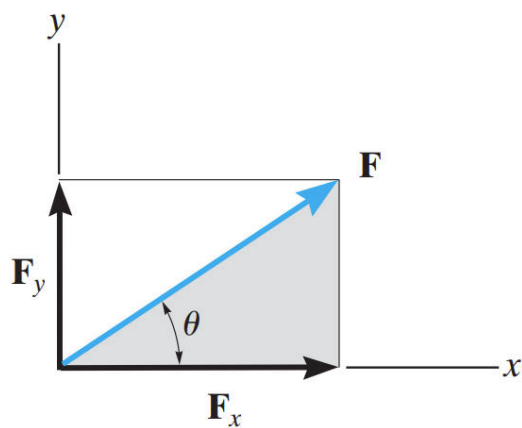
Statics (ME 231)

Lecture 3: Force Vectors (Mathematical Analysis)

By Saeed Lotfan

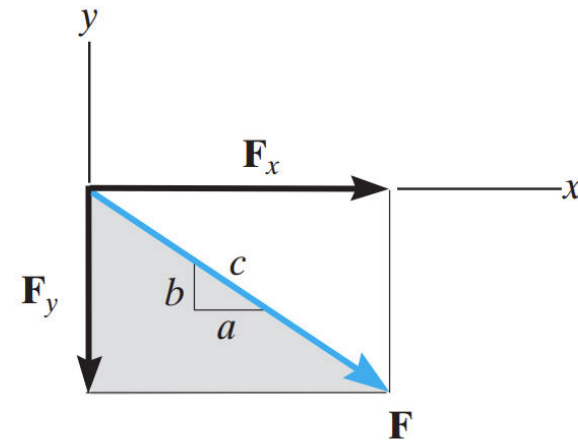
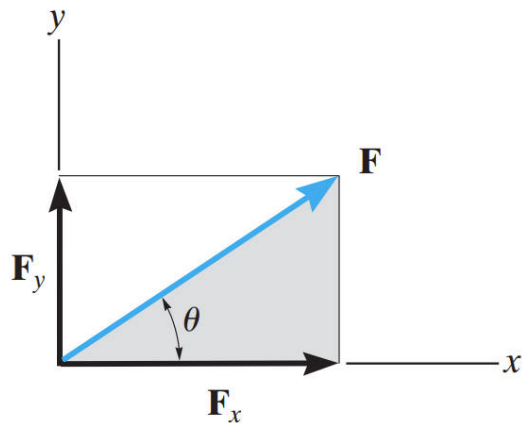
Addition of a System of Coplanar Vectors

When a force is resolved into two components along the x and y axes, the components are then called **rectangular components**.



Addition of a System of Coplanar Vectors

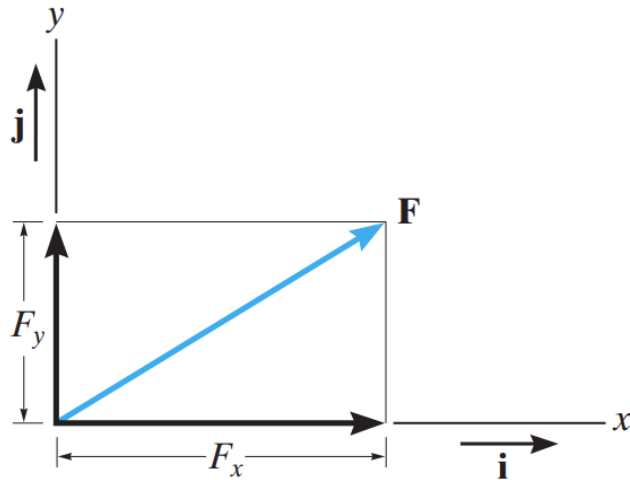
Components: Scalar Notation



$$F_x = F \cos \theta \quad \text{and} \quad F_y = F \sin \theta$$

Addition of a System of Coplanar Vectors

Components: Cartesian Vector Notation

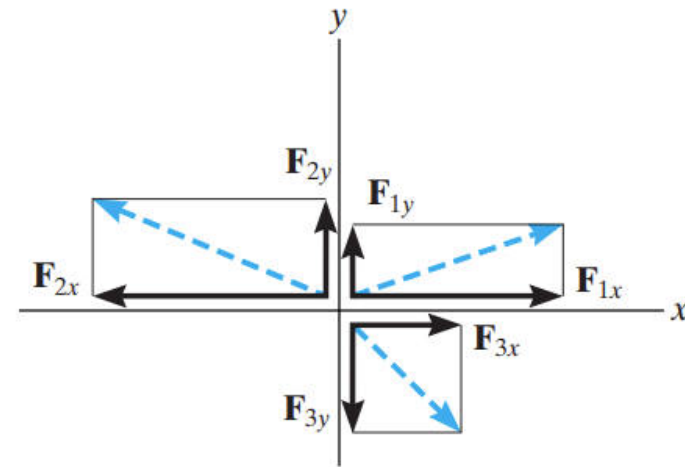
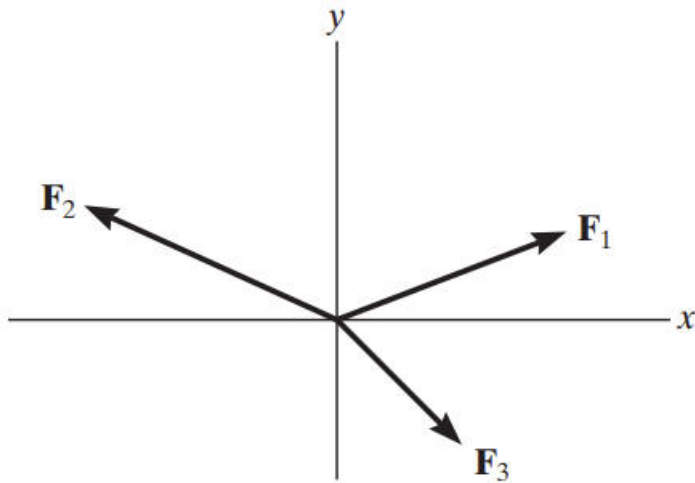


$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j}$$

$$\left\{ \begin{array}{l} |\vec{F}| = \sqrt{F_x^2 + F_y^2} \\ \theta = \tan^{-1} \frac{F_y}{F_x} \end{array} \right.$$

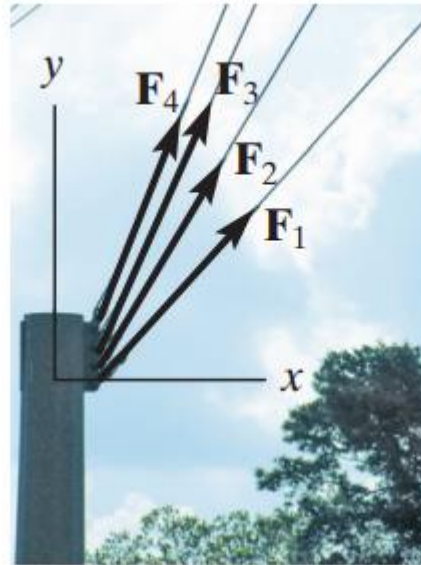
Addition of a System of Coplanar Vectors

Coplanar Force Resultants



$$\begin{aligned}\mathbf{F}_R &= \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 \\ &= F_{1x}\mathbf{i} + F_{1y}\mathbf{j} - F_{2x}\mathbf{i} + F_{2y}\mathbf{j} + F_{3x}\mathbf{i} - F_{3y}\mathbf{j} \\ &= (F_{1x} - F_{2x} + F_{3x})\mathbf{i} + (F_{1y} + F_{2y} - F_{3y})\mathbf{j} \\ &= (F_R)_x\mathbf{i} + (F_R)_y\mathbf{j}\end{aligned}$$

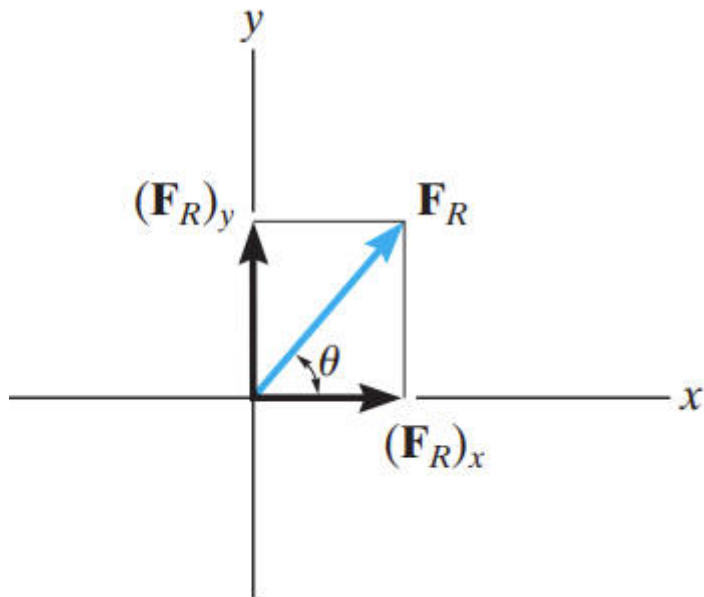
Addition of a System of Coplanar Vectors



The resultant force of the four cable forces acting on the post can be determined by adding algebraically the separate x and y components of each cable force. This resultant $\mathbf{F}_R$ produces the *same pulling effect* on the post as all four cables.

Addition of a System of Coplanar Vectors

In general, the components of the resultant force of any number of coplanar forces can be represented by the algebraic sum of the x and y components of all the forces.



$$(\mathbf{F}_R)_x = \Sigma F_x$$

$$(\mathbf{F}_R)_y = \Sigma F_y$$

$$F_R = \sqrt{(\mathbf{F}_R)_x^2 + (\mathbf{F}_R)_y^2}$$

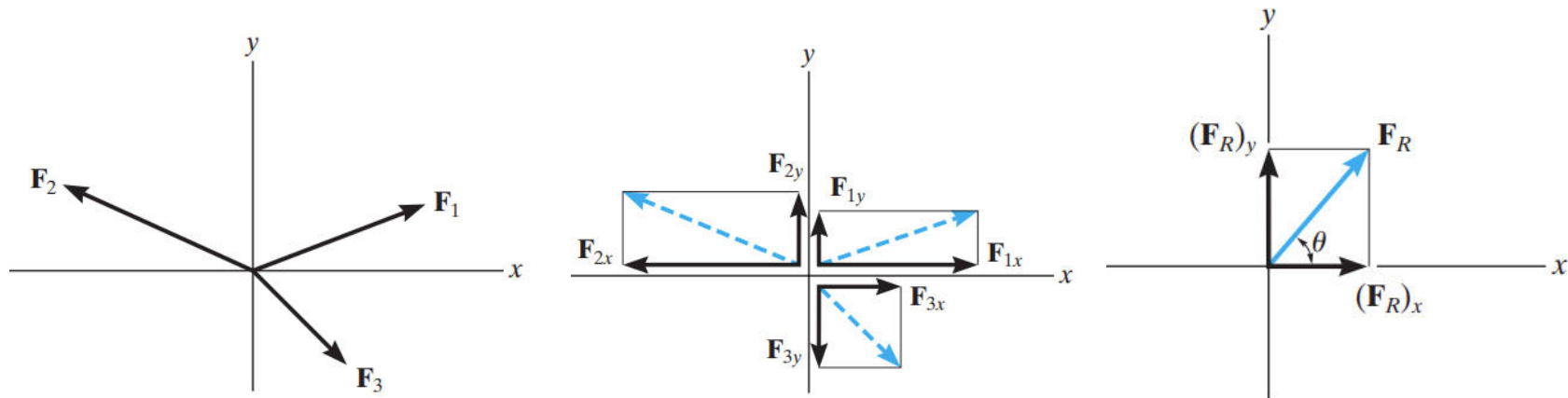
$$\theta = \tan^{-1} \left| \frac{(\mathbf{F}_R)_y}{(\mathbf{F}_R)_x} \right|$$

Concluding Notes

- The resultant of several coplanar forces can easily be determined if an x, y coordinate system is established, and the forces are resolved into components along the axes.
- The direction of each force is specified by the angle its line of action makes with one of the axes, or by a slope triangle.
- The orientation of the x and y axes is arbitrary, and their positive direction can be specified by the Cartesian unit vectors $\mathbf{i}$ and $\mathbf{j}$.
- The x and y components of the resultant force are simply the algebraic addition of the components of all the coplanar forces.

Concluding Notes

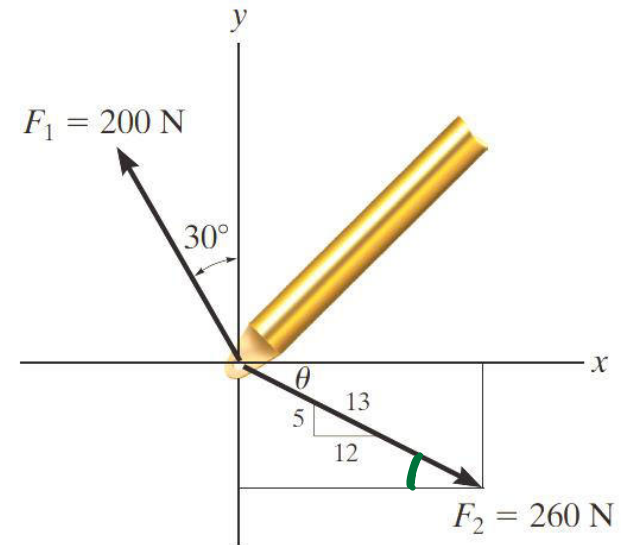
- The magnitude of the resultant force is determined from the Pythagorean theorem, and when the resultant components are sketched on the x and y axes, the direction θ of the resultant can be determined from trigonometry.



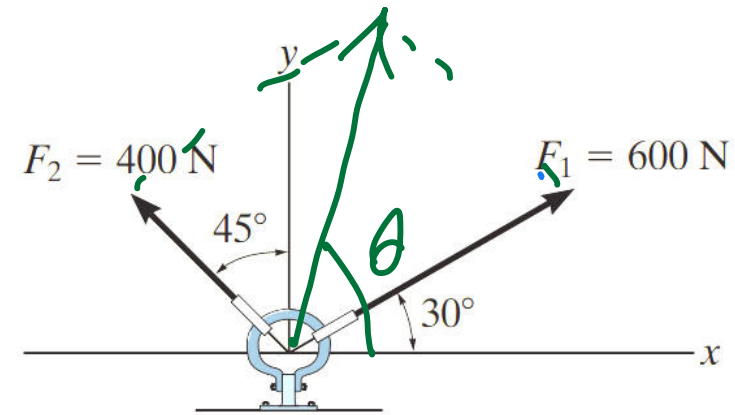
Example: Determine the x and y components of $\mathbf{F}_1$ and $\mathbf{F}_2$ acting on the boom shown in the figure below. Express each force as a Cartesian vector.

$$\vec{F}_1 = -200 \sin 30 \, \underline{\hat{i}} + 200 \cos 30 \, \underline{\hat{j}} \, \text{N}$$

$$\vec{F}_2 = 260 \left(\frac{12}{13} \right) \underline{\hat{i}} - 260 \left(\frac{5}{13} \right) \underline{\hat{j}} \, \text{N}$$



Example: The link shown below is subjected to two forces $\mathbf{F}_1$ and $\mathbf{F}_2$. Determine the magnitude and direction of the resultant force.



$$\vec{F}_1 = 600 \cos 30 \, \underline{\hat{i}} + 600 \sin 30 \, \underline{\hat{j}}$$

$$\vec{F}_2 = -400 \sin 45 \, \underline{\hat{i}} + 400 \cos 45 \, \underline{\hat{j}}$$

$$\vec{F}_R = \underbrace{(600 \cos 30 - 400 \sin 45)}_{F_{Rx}} \, \underline{\hat{i}} + \underbrace{(600 \sin 30 + 400 \cos 45)}_{F_{Ry}} \, \underline{\hat{j}}$$

$$|\vec{F}_R| = \sqrt{F_{Rx}^2 + F_{Ry}^2}$$

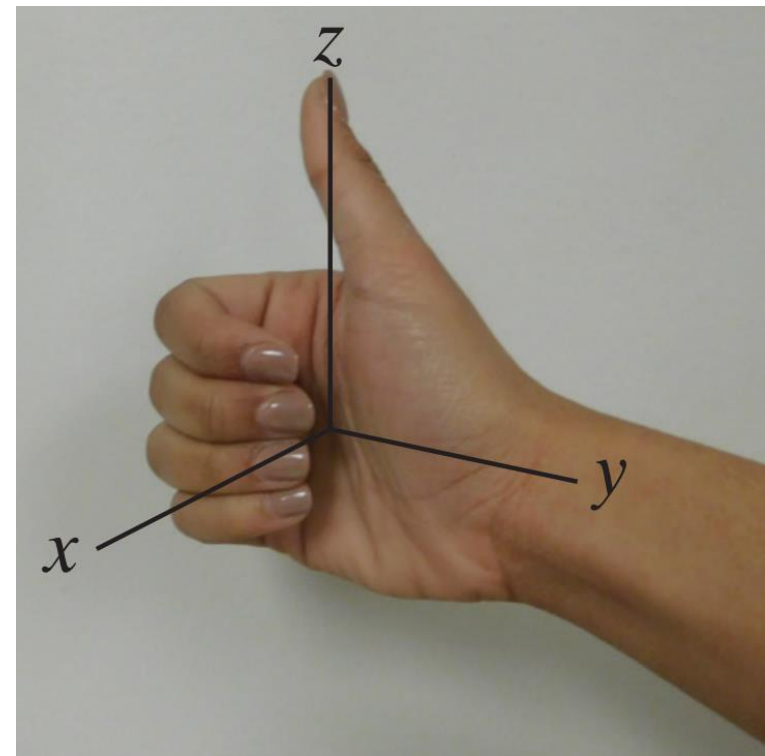
$$\theta = \tan^{-1} \frac{F_{Ry}}{F_{Rx}}$$

Cartesian Vectors

The operations of vector algebra, when applied to solving problems in *three dimensions*, are greatly simplified if the vectors are first represented in Cartesian vector form.

Right-Handed Coordinate System

Specifically, a rectangular coordinate system is said to be right-handed if the thumb of the right hand points in the direction of the positive z axis when the right-hand fingers are curled about this axis and directed from the positive x towards the positive y axis.



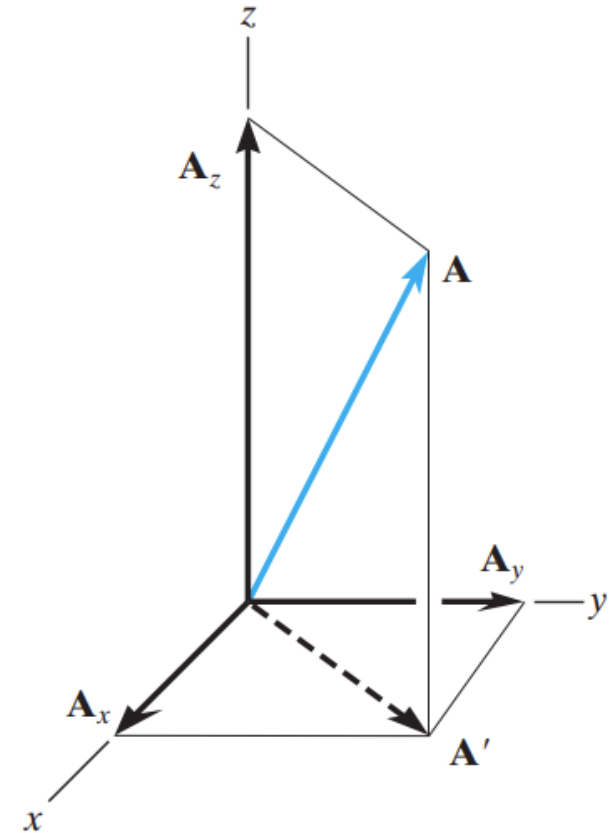
Cartesian Vectors

Rectangular Components of a Vector

$$\mathbf{A} = \mathbf{A}_x + \mathbf{A}_y + \mathbf{A}_z$$

$$\vec{A} = \vec{A}_1 + \vec{A}_2$$

$$\vec{A} = \vec{A}_x + \vec{A}_y + \vec{A}_z$$

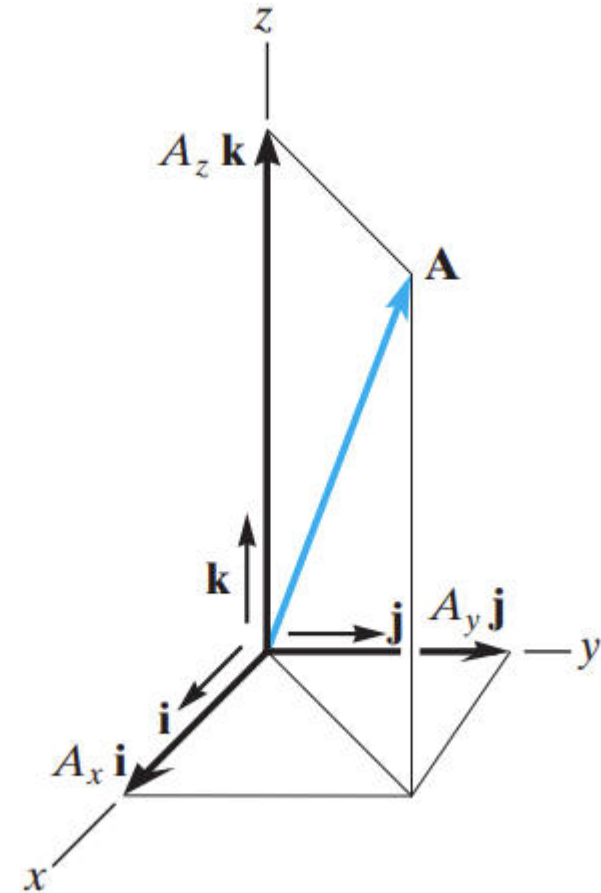


Cartesian Vectors

Cartesian Vector Representation

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$$

There is a distinct advantage to writing vectors in this manner. Separating the *magnitude* and *direction* of each *component vector* will simplify the operations of vector algebra, particularly in three dimensions.

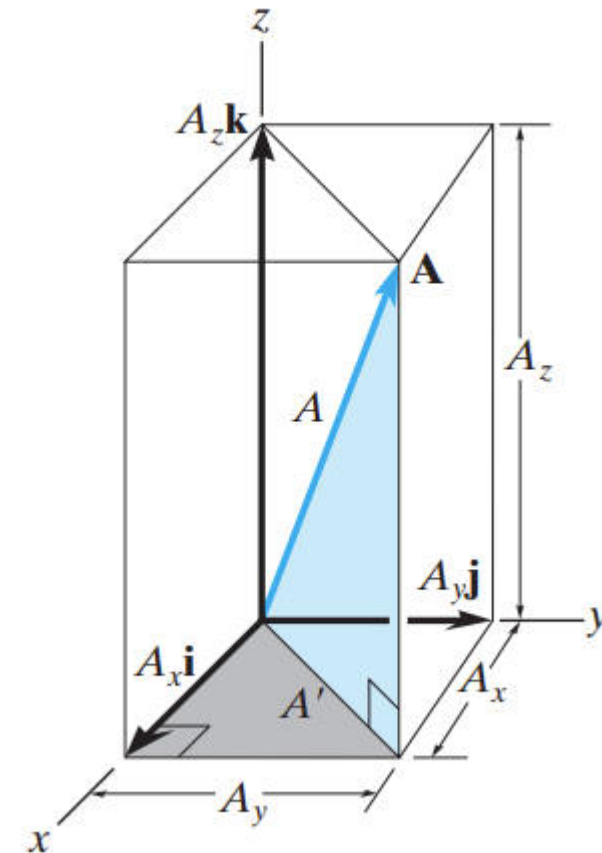


Cartesian Vectors

Magnitude of a Cartesian Vector

The magnitude of **A** is equal to the positive square root of the sum of the squares of the magnitudes of its components.

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$



Cartesian Vectors

Coordinate Direction Angles (*direction cosines*)

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

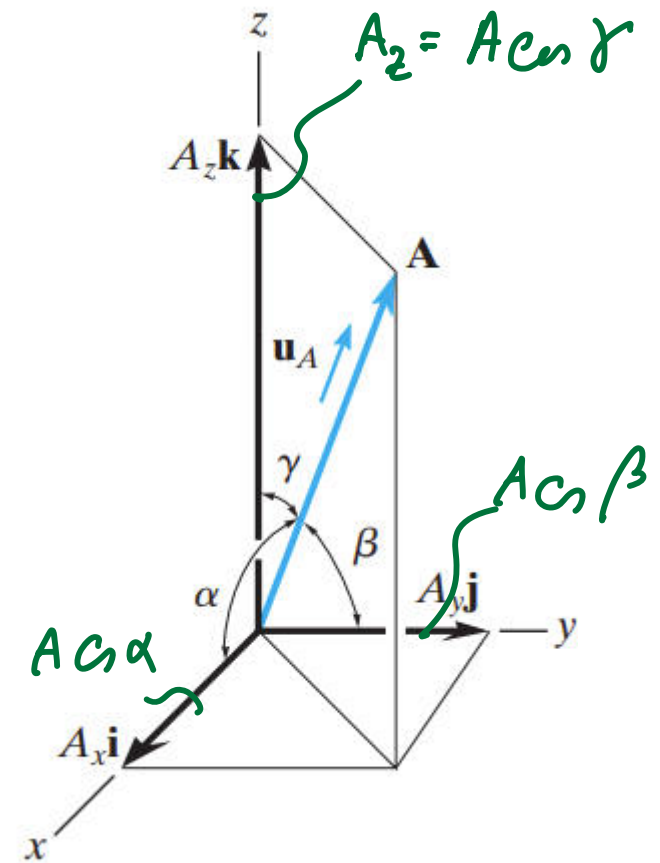
$$\cos \alpha = \frac{A_x}{A} \quad \cos \beta = \frac{A_y}{A} \quad \cos \gamma = \frac{A_z}{A}$$

$$\vec{A} = A_x \underline{\hat{i}} + A_y \underline{\hat{j}} + A_z \underline{\hat{k}}$$

$$\vec{u}_A = \frac{\vec{A}}{|\vec{A}|} = \frac{A_x}{|\vec{A}|} \underline{\hat{i}} + \frac{A_y}{|\vec{A}|} \underline{\hat{j}} + \frac{A_z}{|\vec{A}|} \underline{\hat{k}}$$

$$\vec{u}_A = \frac{|\vec{A}| \cos \alpha}{|\vec{A}|} \underline{\hat{i}} + \frac{|\vec{A}| \cos \beta}{|\vec{A}|} \underline{\hat{j}} + \frac{|\vec{A}| \cos \gamma}{|\vec{A}|} \underline{\hat{k}}$$

$$\vec{u}_A = \cos \alpha \underline{\hat{i}} + \cos \beta \underline{\hat{j}} + \cos \gamma \underline{\hat{k}}$$



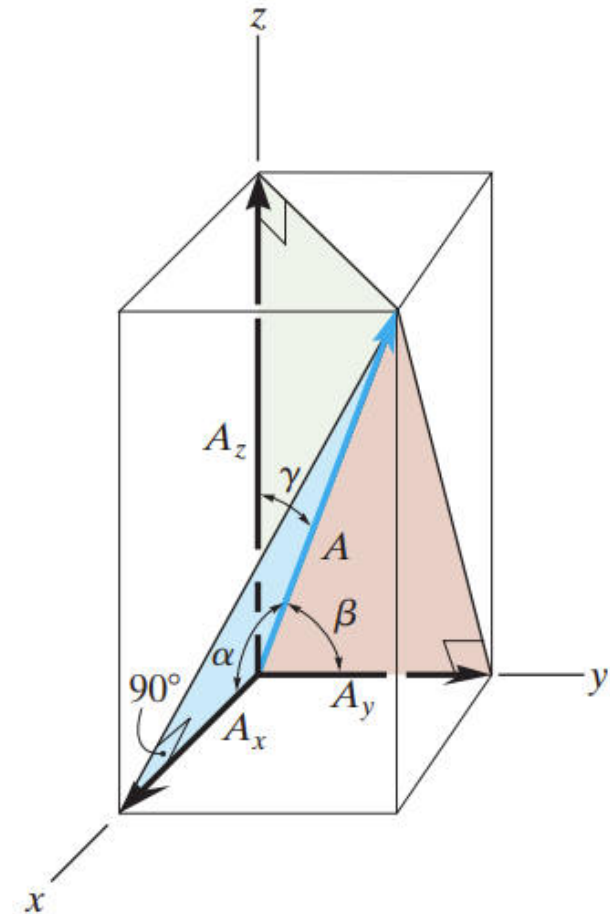
Cartesian Vectors

Direction Cosines and unit vector in the direction of a vector

$$\mathbf{u}_A = \frac{\mathbf{A}}{A} = \frac{A_x}{A}\mathbf{i} + \frac{A_y}{A}\mathbf{j} + \frac{A_z}{A}\mathbf{k}$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\begin{aligned}\mathbf{A} &= A\mathbf{u}_A \\ &= A \cos \alpha \mathbf{i} + A \cos \beta \mathbf{j} + A \cos \gamma \mathbf{k} \\ &= A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}\end{aligned}$$



Cartesian Vectors

Horizontal and Vertical Angles

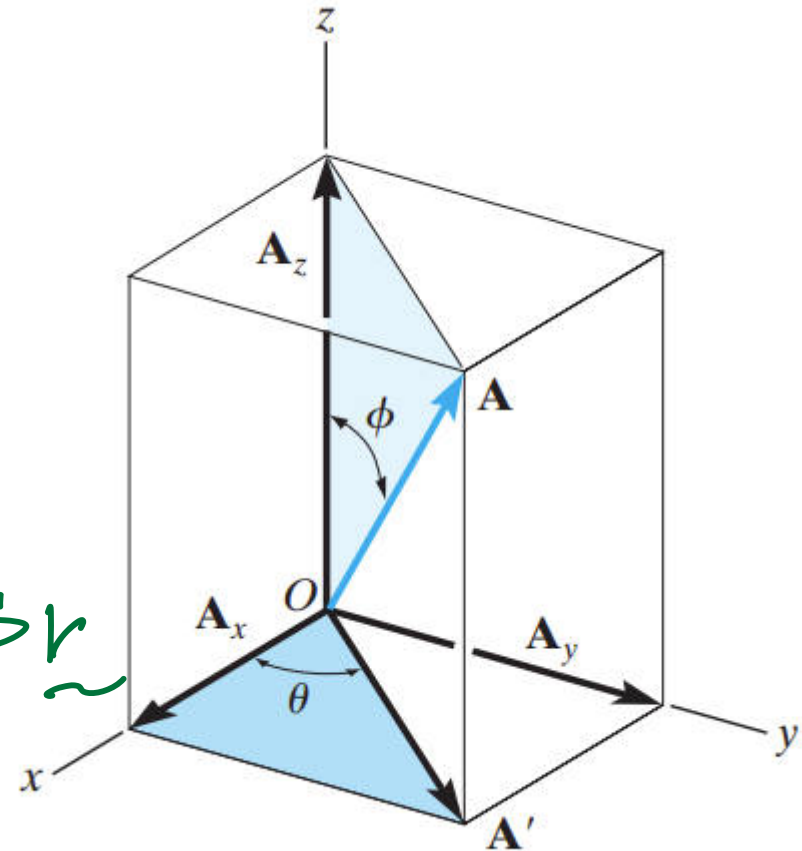
horizontal angle θ

vertical angle ϕ

$$\vec{A} = \vec{A'} + \vec{A_z}$$

$$\vec{A} = A \sin \phi \cos \theta \vec{i} + A \sin \phi \sin \theta \vec{j} + A \cos \phi \vec{k}$$

$$|\vec{A'}| = A \sin \phi$$

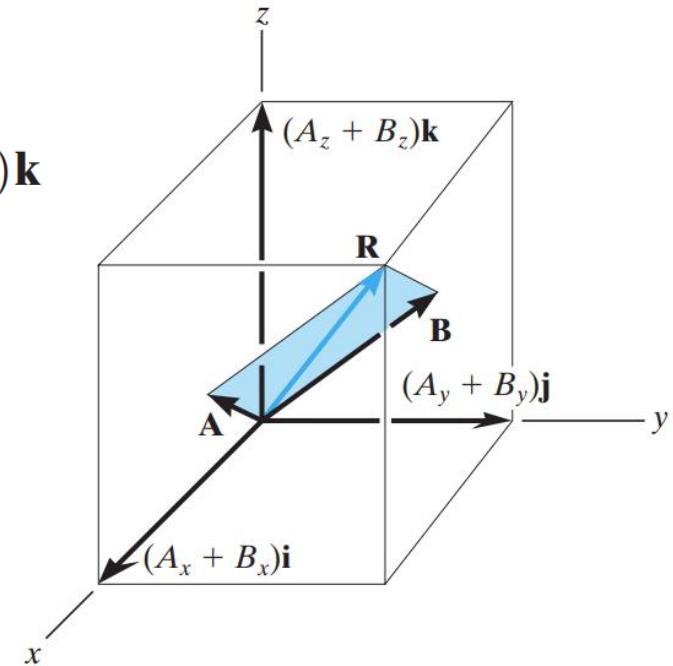


$$\mathbf{A} = A \sin \phi \cos \theta \mathbf{i} + A \sin \phi \sin \theta \mathbf{j} + A \cos \phi \mathbf{k}$$

Addition of Cartesian Vectors

$$\mathbf{R} = \mathbf{A} + \mathbf{B} = (A_x + B_x)\mathbf{i} + (A_y + B_y)\mathbf{j} + (A_z + B_z)\mathbf{k}$$

$$\mathbf{F}_R = \Sigma \mathbf{F} = \Sigma F_x \mathbf{i} + \Sigma F_y \mathbf{j} + \Sigma F_z \mathbf{k}$$

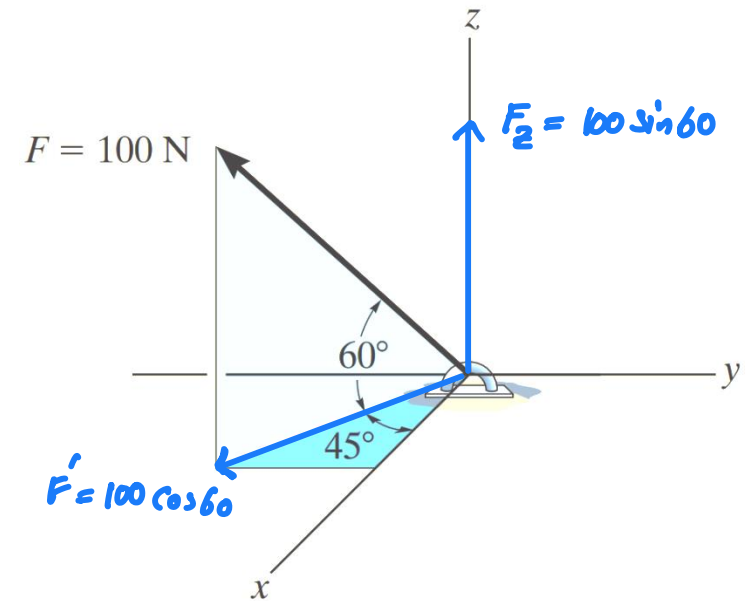


Concluding Notes

- A Cartesian vector **A** has **i**, **j**, **k** components along the *x*, *y*, *z* axes. If **A** is known, its magnitude is $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$
- The direction of a Cartesian vector can be defined by the three coordinate direction angles α , β , γ , measured from the *positive* *x*, *y*, *z* axes to the *tail* of the vector. To find these angles, formulate a unit vector in the direction of **A**, *i.e.*, $\mathbf{u}_A = \mathbf{A}/A$, and determine the inverse cosines of its components. Only two of these angles are independent of one another; the third angle is found from $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.
- The direction of a Cartesian vector can also be specified using a horizontal angle θ and vertical angle ϕ .

Example: Express the force $\mathbf{F}$ shown in below as a Cartesian vector.

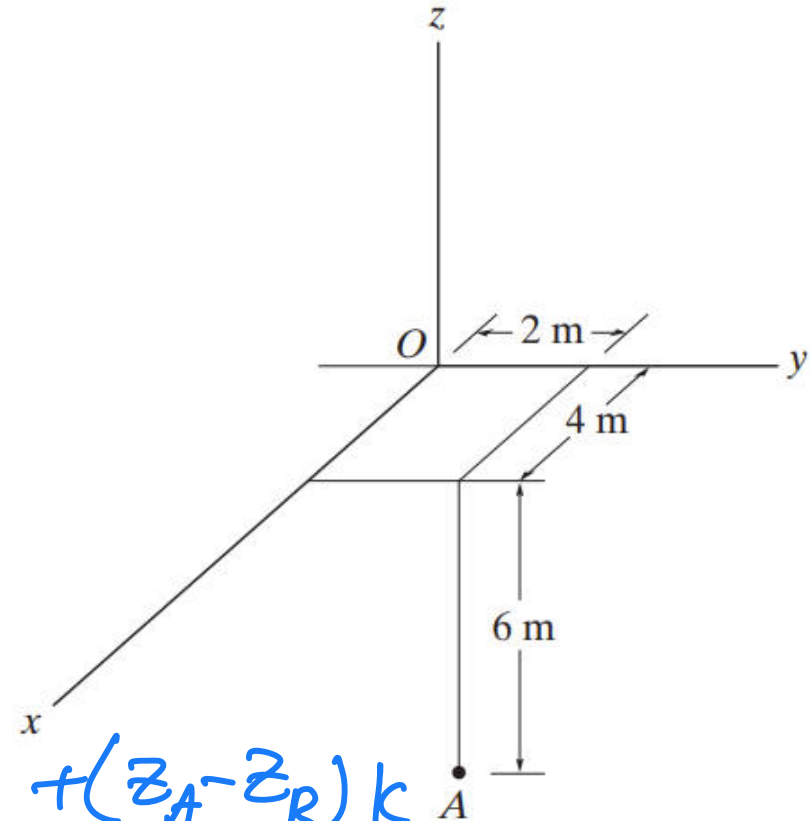
$$\begin{aligned}\vec{F} = & 100 \cos 60^\circ \cos 45^\circ \underline{\hat{i}} \\ & - 100 \cos 60^\circ \sin 45^\circ \underline{\hat{j}} \\ & + 100 \sin 60^\circ \underline{\hat{k}} \text{ N}\end{aligned}$$



Position Vectors

x, y, z Coordinates

Generally, most of the time, the positive z axis to be directed upward (the zenith direction) so that it measures the height of an object or the altitude of a point. The x, y axes then lie in the horizontal plane.

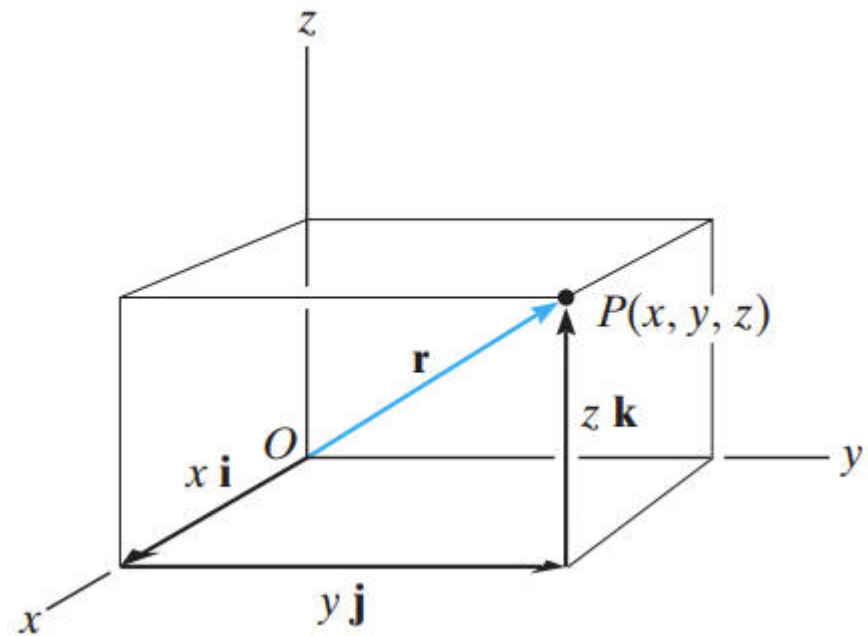
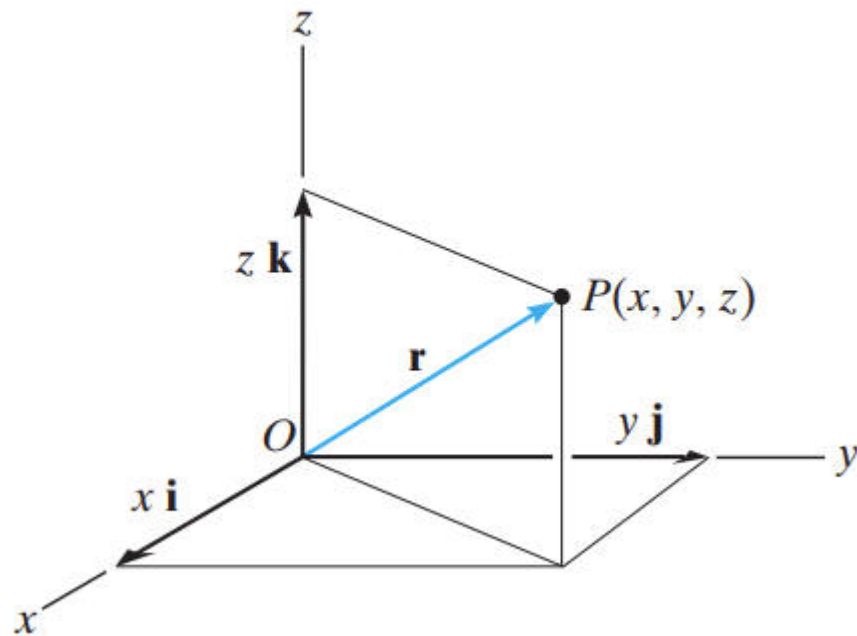


$$\vec{r}_{A/O} = 4\hat{i} + 2\hat{j} - 6\hat{k} \text{ m}$$

$$\vec{r}_{A/B} = (x_A - x_B)\hat{i} + (y_A - y_B)\hat{j} + (z_A - z_B)\hat{k}$$

Position Vectors

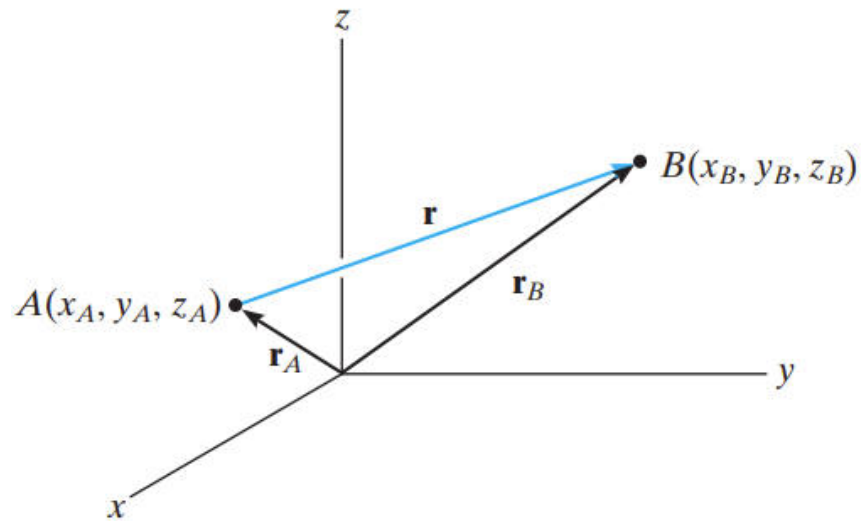
A position vector $\mathbf{r}$ is defined as a fixed vector which locates a point in space relative to another point.



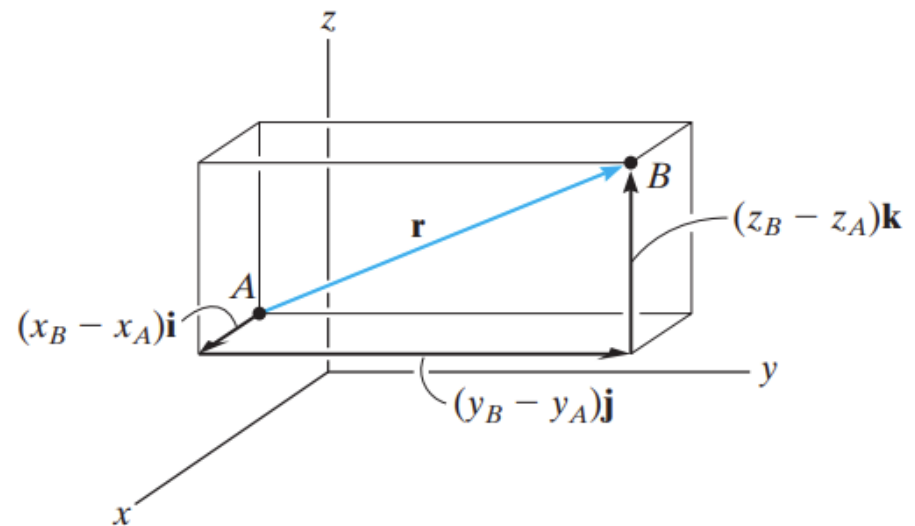
$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

Position Vectors

General Case



$$\mathbf{r}_A + \mathbf{r} = \mathbf{r}_B$$



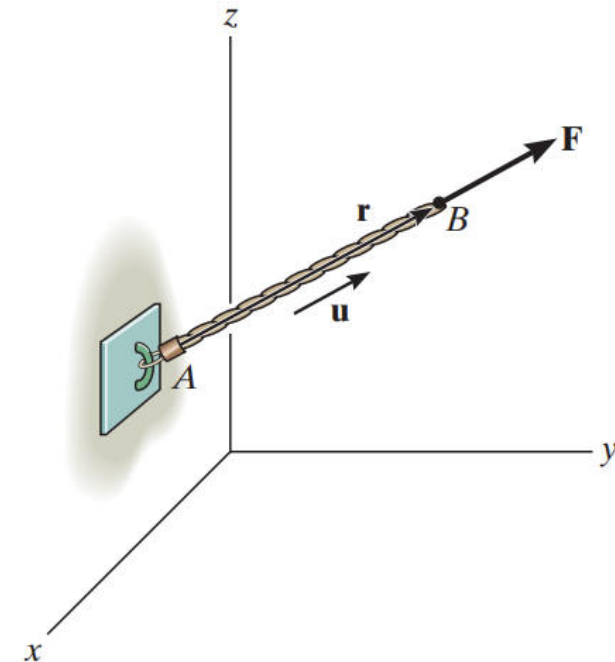
$$\mathbf{r} = \mathbf{r}_B - \mathbf{r}_A = (x_B\mathbf{i} + y_B\mathbf{j} + z_B\mathbf{k}) - (x_A\mathbf{i} + y_A\mathbf{j} + z_A\mathbf{k})$$



If an x, y, z coordinate system is established, then the coordinates of two points A and B on the cable can be determined. From this the position vector $\mathbf{r}$ acting along the cable can be formulated. Its magnitude represents the distance from A to B , and its unit vector, $\mathbf{u} = \mathbf{r}/r$, gives the direction defined by α, β, γ .

Force Vectors Directed Along a Line

Quite often in three-dimensional statics problems, the direction of a force is specified by two points through which its line of action passes. Such a situation is shown in this figure, where force $\mathbf{F}$ is directed along the cord AB. We can formulate $\mathbf{F}$ as a Cartesian vector by realizing that it has the same direction and sense as the position vector $\mathbf{r}$ directed from point A to point B on the cord.



$$\mathbf{F} = F \mathbf{u} = F \left(\frac{\mathbf{r}}{r} \right) = F \left(\frac{(x_B - x_A)\mathbf{i} + (y_B - y_A)\mathbf{j} + (z_B - z_A)\mathbf{k}}{\sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}} \right)$$



The force $\mathbf{F}$ acting along the rope can be represented as a Cartesian vector by establishing x, y, z axes and first forming a position vector $\mathbf{r}$ along the length of the rope. Then the corresponding unit vector $\mathbf{u} = \mathbf{r}/r$ that defines the direction of both the rope and the force can be determined. Finally, the magnitude of the force is combined with its direction, so that $\mathbf{F} = Fu$.

Concluding Notes

- A position vector locates one point in space relative to another point.
- The easiest way to formulate the components of a position vector is to determine the distance and direction that one must travel along the x , y , z directions—going from the tail to the head of the vector.
- A force $\mathbf{F}$ acting in the direction of a position vector $\mathbf{r}$ can be represented in Cartesian form if the unit vector $\mathbf{u}$ of the position vector is determined and it is multiplied by the magnitude of the force, i.e., $\mathbf{F} = F\mathbf{u} = F(\mathbf{r}/r)$.

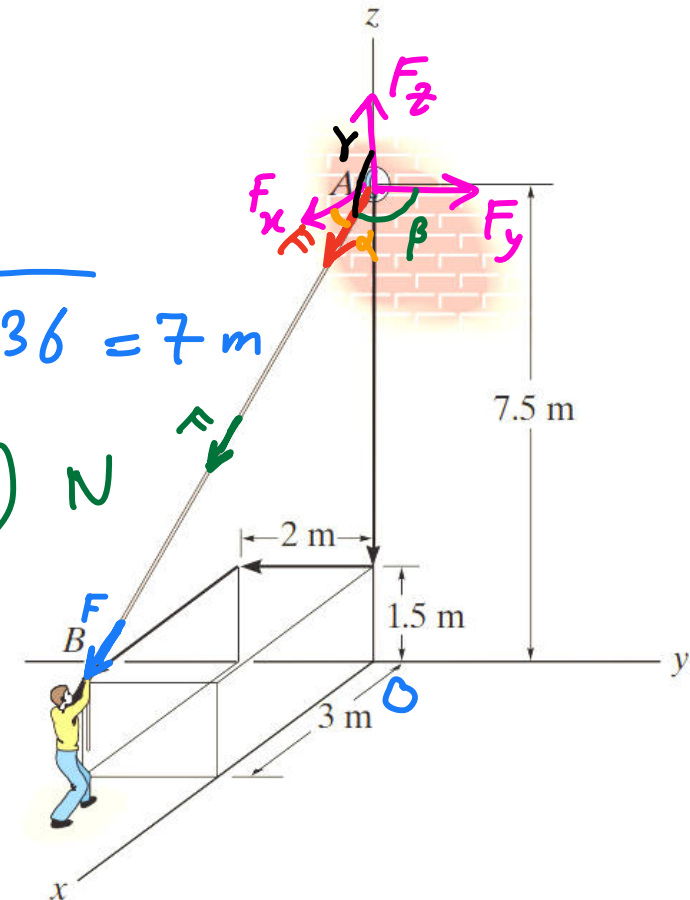
Example: The man shown in figure below pulls on the cord with a force of 350 N. Represent this force acting on the support A as a Cartesian vector and determine its direction.

$$\begin{aligned}\vec{r}_{B/A} &= \vec{r}_{B/O} - \vec{r}_{A/O} \\ &= (3\hat{i} - 2\hat{j} + 1.5\hat{k}) - (7.5\hat{k}) \text{ m} \\ \vec{r}_{B/A} &= 3\hat{i} - 2\hat{j} - 6\hat{k} \text{ m} \quad |\vec{r}_{B/A}| = \sqrt{9 + 4 + 36} = 7 \text{ m}\end{aligned}$$

$$\vec{F} = 350 \vec{u}_F = 350 \left(\frac{3}{7}\hat{i} - \frac{2}{7}\hat{j} - \frac{6}{7}\hat{k} \right) \text{ N}$$

$$F_x = 150 \text{ N} \rightarrow, \quad F_y = 100 \text{ N} \leftarrow, \quad F_z = 300 \text{ N} \downarrow$$

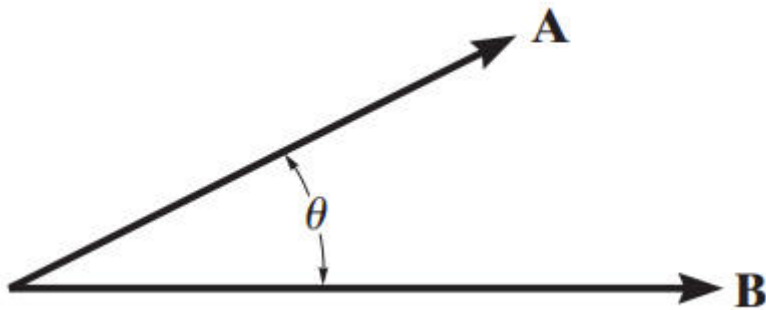
$$\cos \alpha = \frac{3}{7} \quad \cos \beta = -\frac{2}{7} \quad \cos \gamma = -\frac{6}{7}$$



Dot Product

Definition

The dot product of vectors **A** and **B**, written **A**·**B** and read “**A** dot **B**,” is defined as the product of the magnitudes of **A** and **B** and the cosine of the angle θ between their tails.



$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$$

Dot Product

Laws of Operation

1. Commutative law: $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
2. Multiplication by a scalar: $a (\mathbf{A} \cdot \mathbf{B}) = (a\mathbf{A}) \cdot \mathbf{B} = \mathbf{A} \cdot (a\mathbf{B})$
3. Distributive law: $\mathbf{A} \cdot (\mathbf{B} + \mathbf{D}) = (\mathbf{A} \cdot \mathbf{B}) + (\mathbf{A} \cdot \mathbf{D})$

Dot Product

Cartesian Vector Formulation

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= (A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}) \cdot (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}) \\ &= A_x B_x (\mathbf{i} \cdot \mathbf{i}) + A_x B_y (\mathbf{i} \cdot \mathbf{j}) + A_x B_z (\mathbf{i} \cdot \mathbf{k}) \\ &\quad + A_y B_x (\mathbf{j} \cdot \mathbf{i}) + A_y B_y (\mathbf{j} \cdot \mathbf{j}) + A_y B_z (\mathbf{j} \cdot \mathbf{k}) \\ &\quad + A_z B_x (\mathbf{k} \cdot \mathbf{i}) + A_z B_y (\mathbf{k} \cdot \mathbf{j}) + A_z B_z (\mathbf{k} \cdot \mathbf{k})\end{aligned}$$

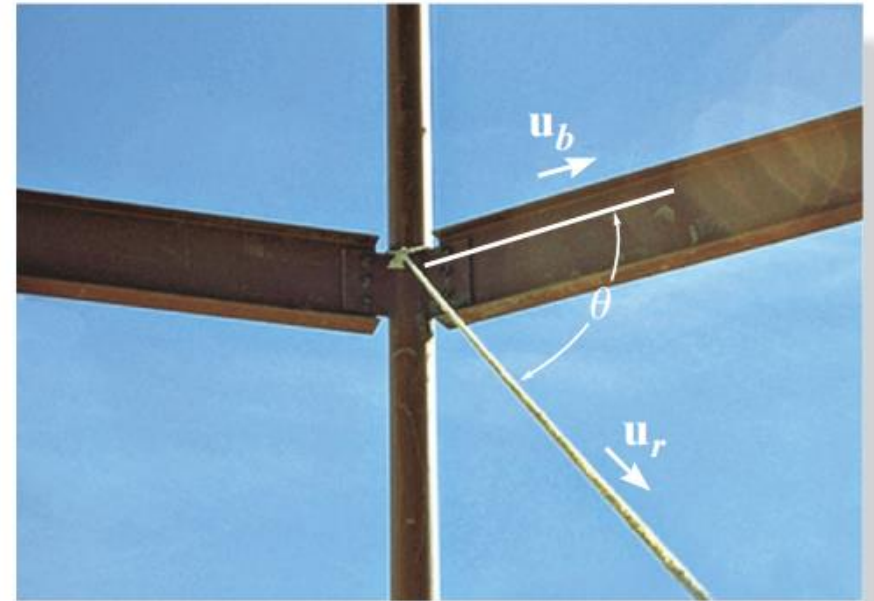
$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$$

Dot Product

Applications: *The angle formed between two vectors or intersecting lines*

$$\theta = \cos^{-1}\left(\frac{\mathbf{A} \cdot \mathbf{B}}{AB}\right) \quad 0^\circ \leq \theta \leq 180^\circ$$

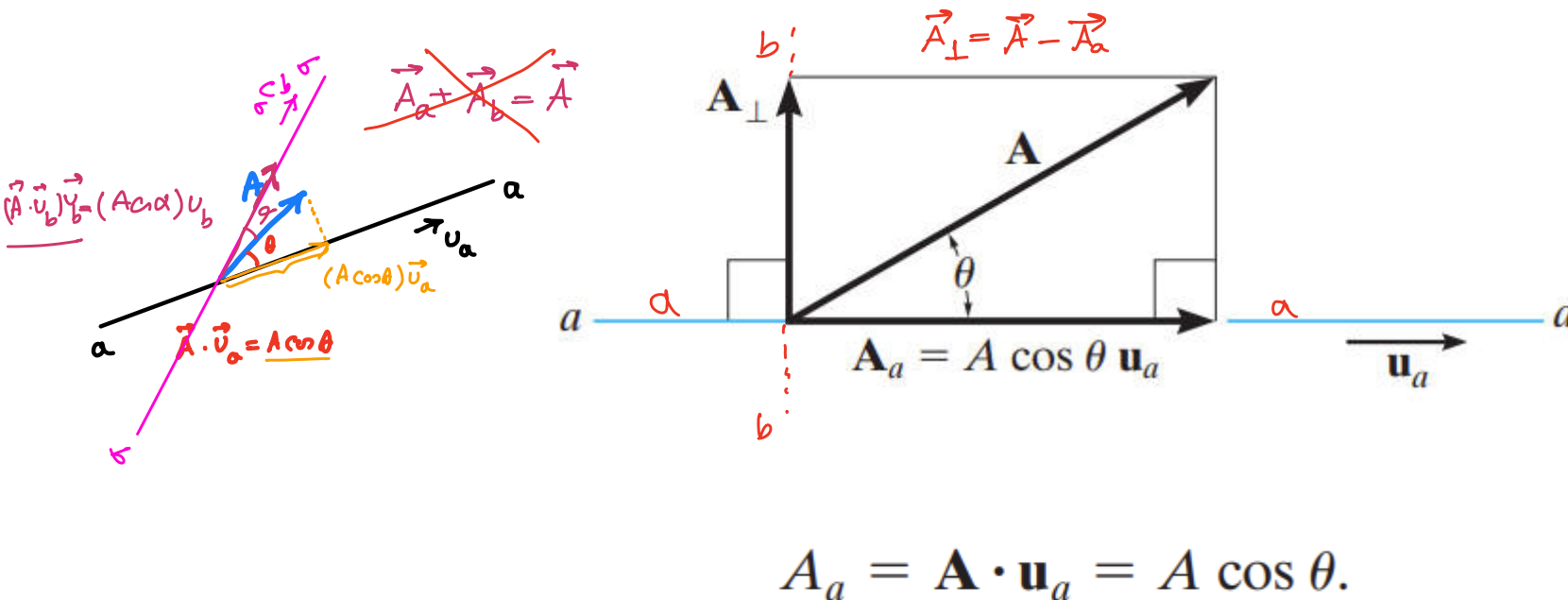
Therefore, if $\mathbf{A} \cdot \mathbf{B} = 0$,
then $\theta = \cos^{-1}(0) = 90^\circ$
so that $\mathbf{A}$ will be perpendicular to $\mathbf{B}$.



The angle θ between the rope and the beam can be determined by formulating unit vectors along the beam and rope and then using the dot product $\mathbf{u}_b \cdot \mathbf{u}_r = (1)(1) \cos \theta$.

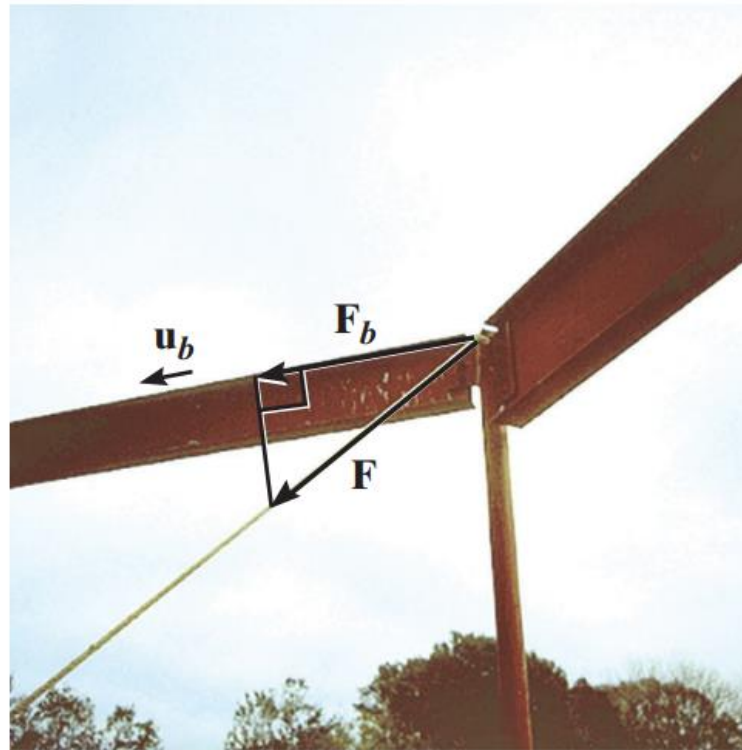
Dot Product

Applications: *The components of a vector parallel and perpendicular to a line*



This component is sometimes referred to as the **projection** of $\mathbf{A}$ onto the line, since a right angle is formed in the construction.

Dot Product

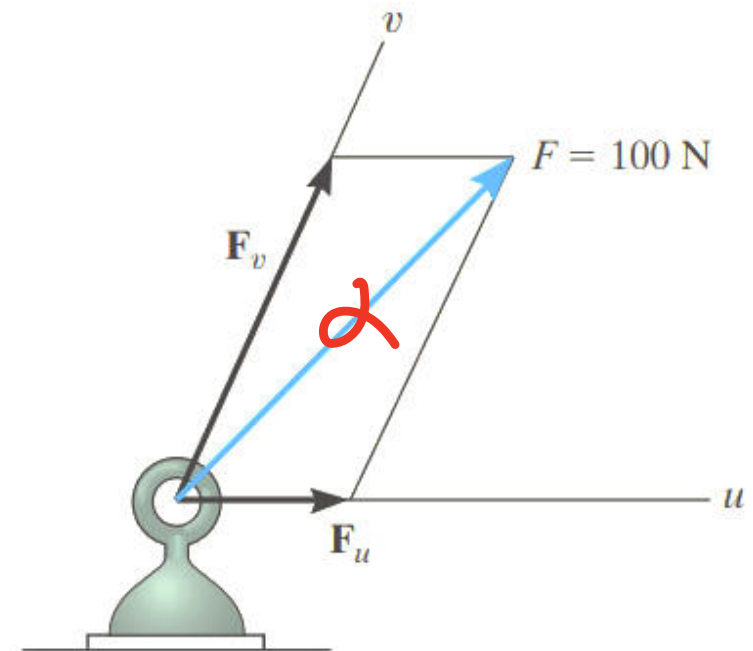
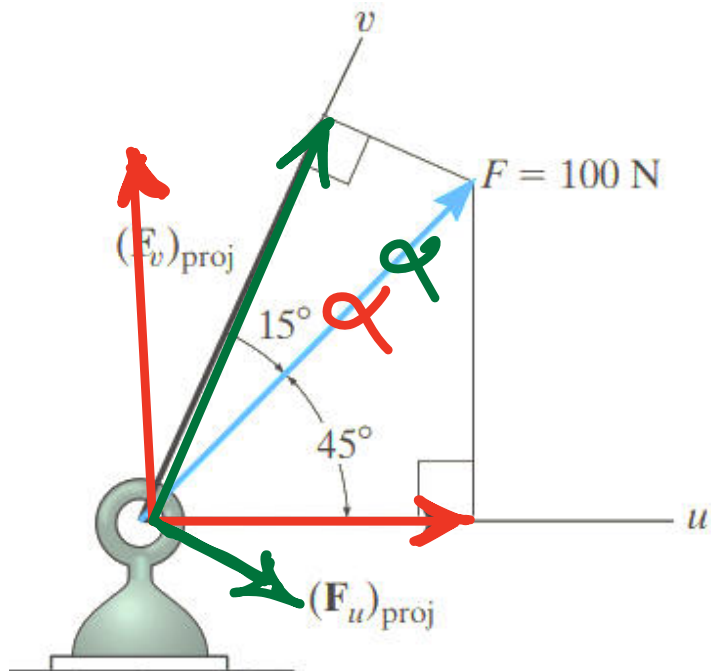


The projection of the cable force $\mathbf{F}$ along the beam can be determined by first finding the unit vector $\mathbf{u}_b$ that defines this direction. Then apply the dot product, $F_b = \mathbf{F} \cdot \mathbf{u}_b$.

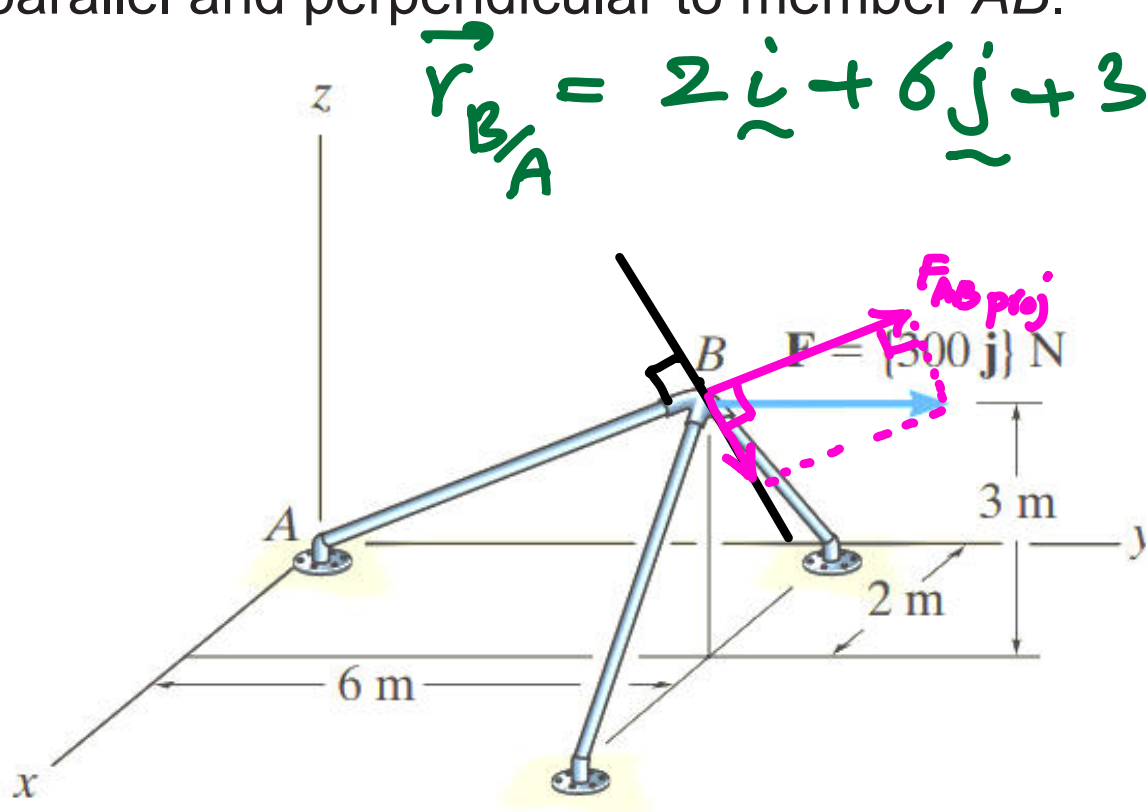
Concluding Notes

- The dot product is used to determine the angle between two vectors or the projection of a vector in a specified direction.
- If vectors **A** and **B** are expressed in Cartesian vector form, the dot product is determined by multiplying the respective x , y , z scalar components and algebraically adding the results, i.e., $\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$.
- From the definition of the dot product, the angle formed between the tails of vectors **A** and **B** is $\theta = \cos^{-1}(\mathbf{A} \cdot \mathbf{B}/AB)$.
- The magnitude of the projection of vector **A** along a line aa whose direction is specified by $\mathbf{u}_a$ is determined from $A_a = \mathbf{A} \cdot \mathbf{u}_a$.

Example: Determine the magnitudes of the projections of the force $\mathbf{F}$ onto the u and v axes.



Example: The frame shown below is subjected to a horizontal force $\mathbf{F} = \{300\mathbf{j}\}$ N. Determine the magnitudes of the components of this force parallel and perpendicular to member AB.



$$\vec{r}_{B/A} = 2\hat{i} + 6\hat{j} + 3\hat{k} \text{ m} \rightarrow \vec{u}_{AB} = \frac{2}{7}\hat{i} + \frac{6}{7}\hat{j} + \frac{3}{7}\hat{k}$$

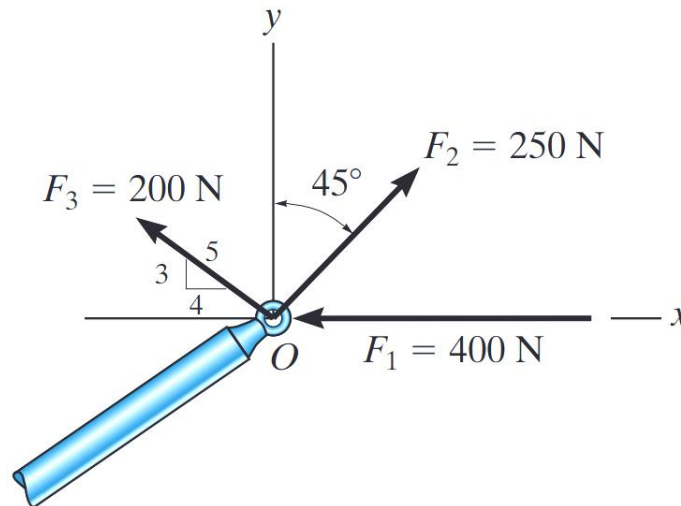
$$\vec{F}_{\perp AB} + (\vec{F}_{AB})_{\text{proj}} = \vec{F}$$

$$F_{\perp} = F - F_{AB \text{ proj}}$$

$$(\vec{F}_{AB})_{\text{proj}} = (\vec{F} \cdot \vec{u}_{AB}) \vec{u}_{AB} = \frac{300 \times 6}{7} \left(\frac{2}{7}\hat{i} + \frac{6}{7}\hat{j} + \frac{3}{7}\hat{k} \right) \text{ N}$$

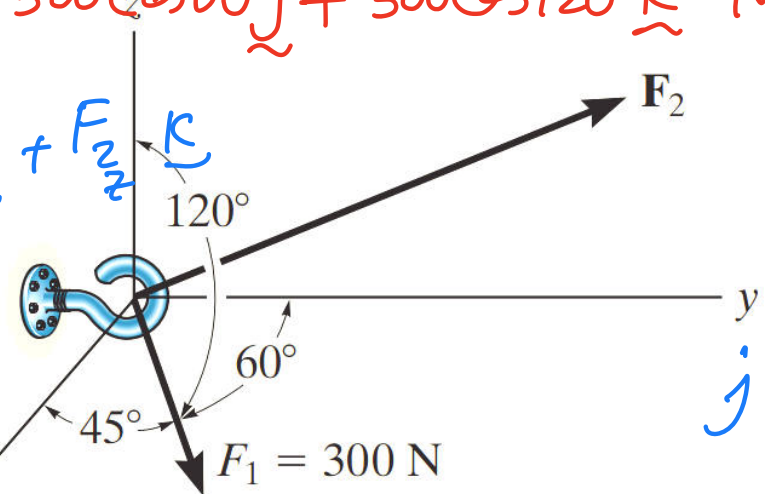
Practice at Home

1. The end of the boom O shown below is subjected to three concurrent and coplanar forces. Determine the magnitude and direction of the resultant force.



Practice at Home

2. Two forces act on the hook shown below. Specify the magnitude of $\mathbf{F}_2$ and its coordinate direction angles so that the resultant force $\mathbf{F}_R$ acts along the positive y axis and has a magnitude of 800 N.



Handwritten solutions:

$$\vec{F}_1 = 300 \cos 45^\circ \underline{i} + 300 \cos 60^\circ \underline{j} + 300 \cos 120^\circ \underline{k} \text{ N}$$

$$\vec{F}_2 = F_{2x} \underline{i} + F_{2y} \underline{j} + F_{2z} \underline{k}$$

$$\vec{F}_R = 800 \underline{j}$$

$$\vec{F}_R = \vec{F}_1 + \vec{F}_2$$

Component equations:

$$i: 300 \cos 45^\circ + F_{2x} = 0$$

$$j: 300 \cos 60^\circ + F_{2y} = 800$$

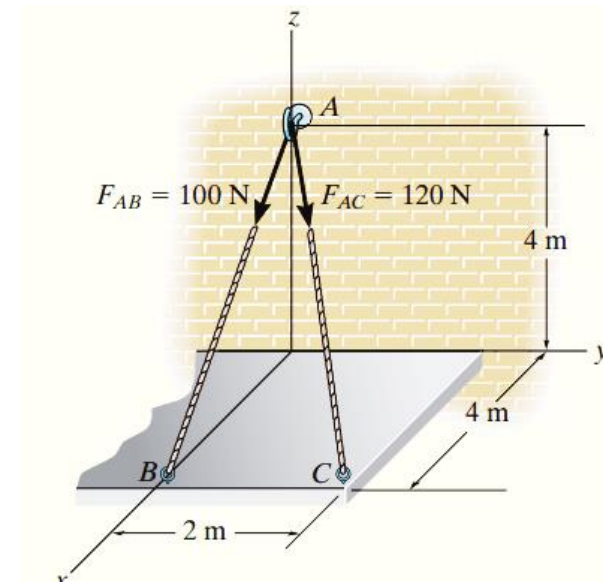
$$k: 300 \cos 120^\circ + F_{2z} = 0$$

$$|F_2| = \sqrt{F_{2x}^2 + F_{2y}^2 + F_{2z}^2}$$

$$u_{F_2} = \frac{\vec{F}_2}{|F_2|} = \cos \alpha \underline{i} + \cos \beta \underline{j} + \cos \gamma \underline{k}$$

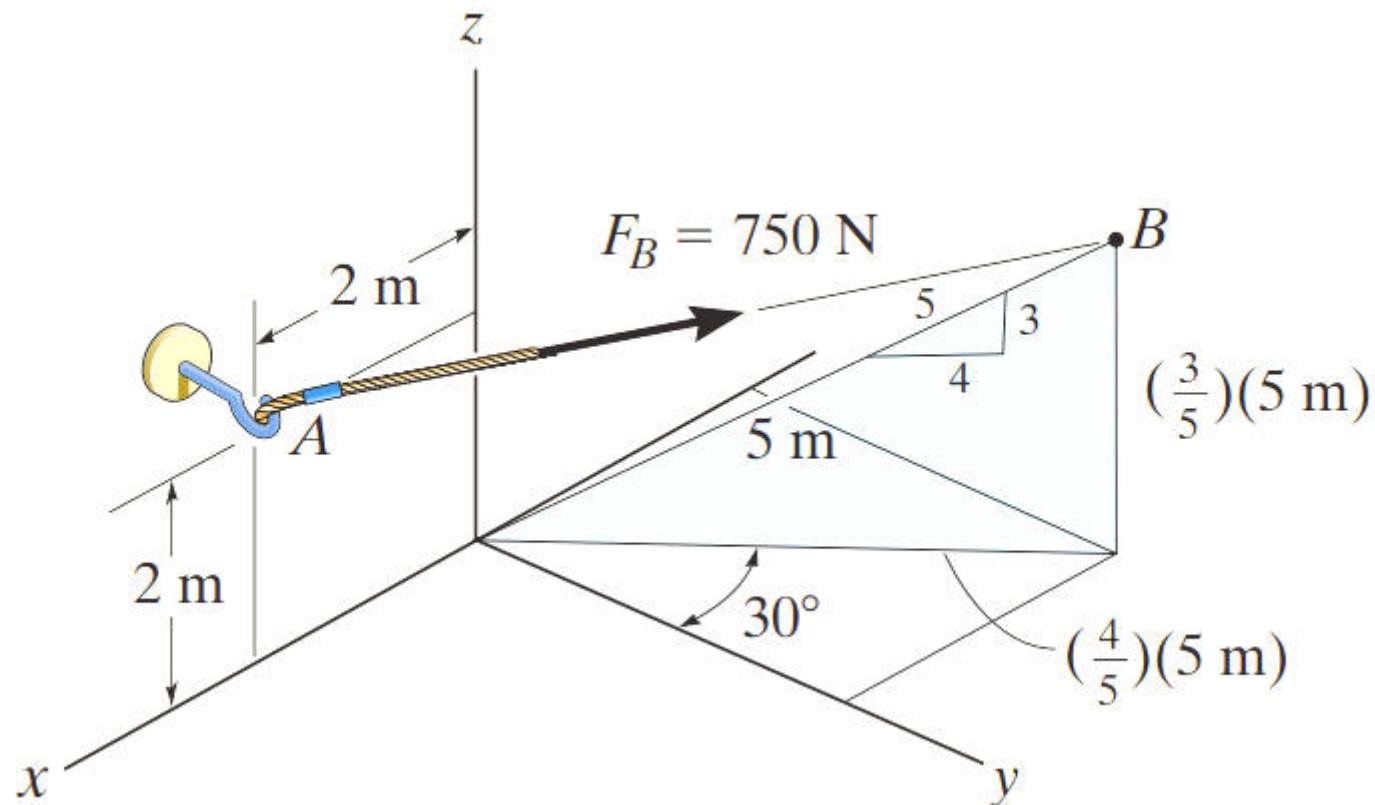
Practice at Home

3. The roof is supported by two cables as shown in the photo. If the cables exert forces $F_{AB} = 100 \text{ N}$ and $F_{AC} = 120 \text{ N}$ on the wall hook at A , determine the resultant force acting at A . Express the result as a Cartesian vector.



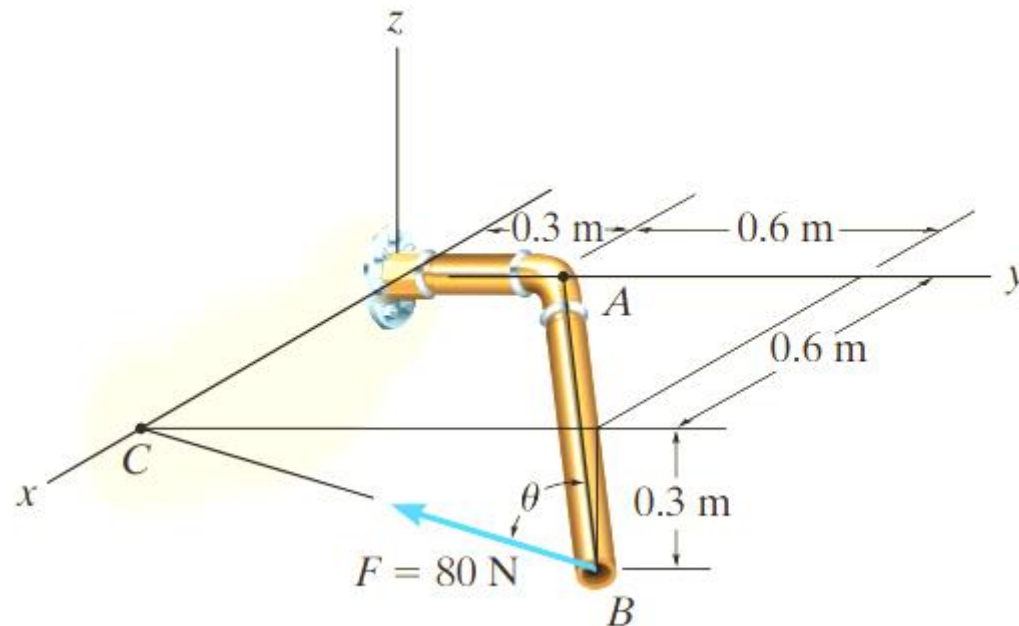
Practice at Home

4. The force shown below acts on the hook. Express it as a Cartesian vector.



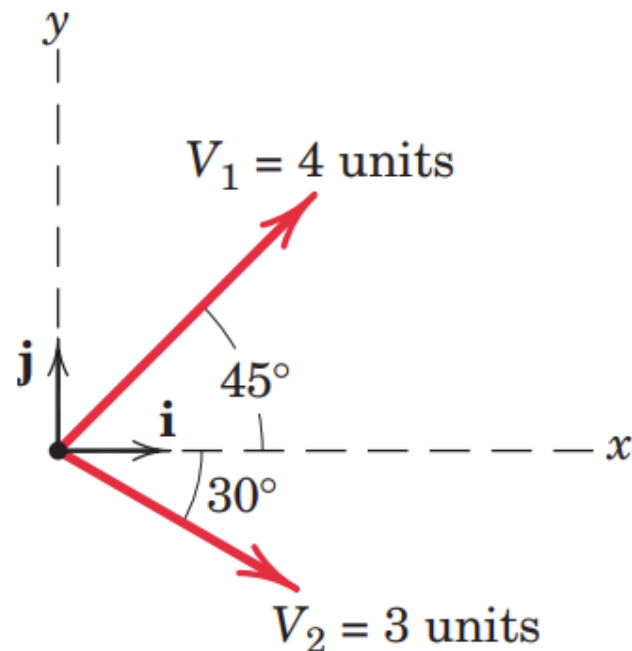
Practice at Home

5. The pipe in figure below is subjected to the force of $F = 80$ N. Determine the angle θ between $\mathbf{F}$ and the pipe segment BA and the projection of $\mathbf{F}$ along this segment.



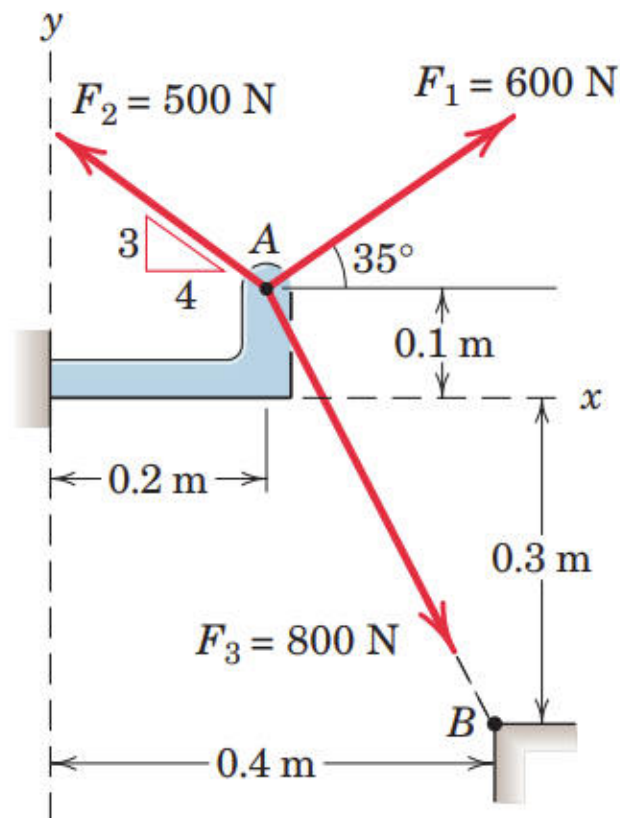
Practice at Home

6. For the vectors $\mathbf{V}_1$ and $\mathbf{V}_2$ shown in the figure, (a) determine the magnitude $\mathbf{S}$ of their vector sum, (b) determine the angle α between $\mathbf{S}$ and the positive x-axis, (c) write $\mathbf{S}$ as a vector in terms of the unit vectors $\mathbf{i}$ and $\mathbf{j}$ and then write a unit vector $\mathbf{n}$ along the vector sum $\mathbf{S}$, finally (d) determine the vector difference $\mathbf{D} = \mathbf{V}_1 - \mathbf{V}_2$.



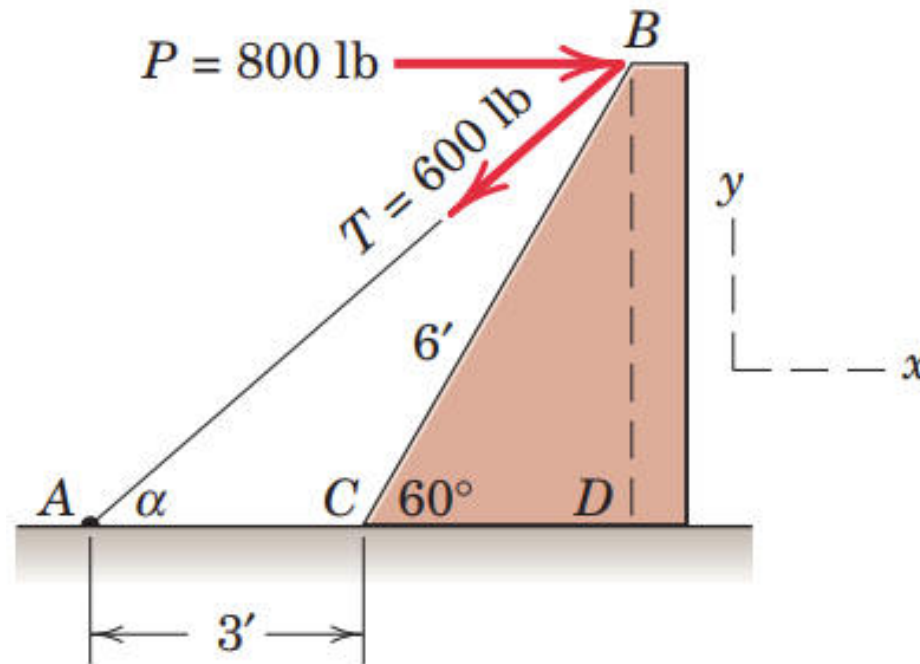
Practice at Home

7. The forces $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$, all of which act on point A of the bracket, are specified in three different ways. Determine the x and y scalar components of each of the three forces.



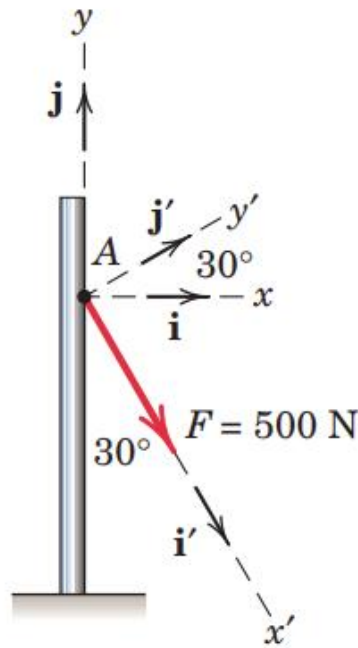
Practice at Home

8. Combine the two forces **P** and **T**, which act on the fixed structure at B, into a single equivalent force **R**.



Practice at Home

9. The 500-N force $\mathbf{F}$ is applied to the vertical pole as shown. (1) Write $\mathbf{F}$ in terms of the unit vectors $\mathbf{i}$ and $\mathbf{j}$ and identify both its vector and scalar components. (2) Determine the scalar components of the force vector $\mathbf{F}$ along the x' - and y' -axes. (3) Determine the scalar components of $\mathbf{F}$ along the x - and y -axes.



Practice at Home

10. Forces $\mathbf{F}_1$ and $\mathbf{F}_2$ act on the bracket as shown. Determine the projection $\mathbf{F}_b$ of their resultant $\mathbf{R}$ onto the b -axis.

